Homework 4

Zara Mecklai

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```r
if (!require("pacman")) install.packages("pacman")
```

```
Loading required package: pacman
```

```r
pacman::p_load(tidyverse, ggplot2, dplyr, lubridate, ivreg, modelsummary)
```

```r
final.hcris.data <- read_csv('../data/output/HCRIS_final.csv')
```

```
Rows: 133679 Columns: 51
-- Column specification --------------------------------------------------------
Delimiter: ","
chr   (9): provider_number, data_source, street, city, state, zip, county, n...
dbl  (38): beds, tot_charges, net_pat_rev, tot_discounts, tot_operating_exp,...
date  (4): fy_start, fy_end, date_processed, date_created

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#Q1
final.hcris.v1996 = read_csv('../data/output/HCRIS_Data_v1996.csv')
```

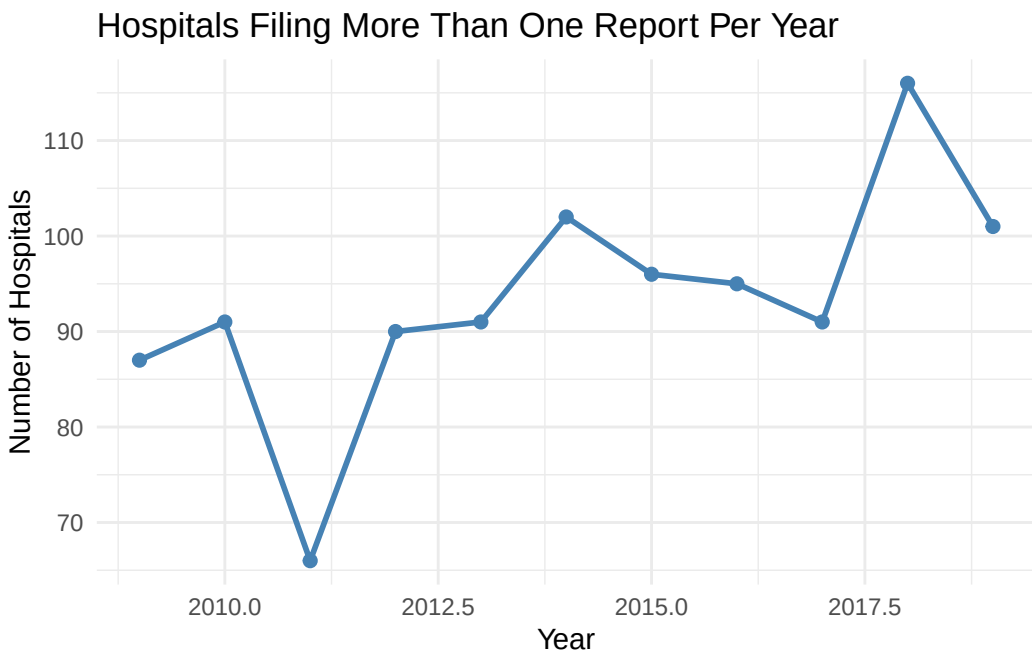
```
Rows: 78508 Columns: 47
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (12): provider_number, fy_start, fy_end, date_processed, date_created, d...
dbl (34): report, status, year, beds, tot_charges, net_pat_rev, tot_discount...
lgl  (1): npi

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
final.hcris.v2010 = read_csv('../data/output/HCRIS_Data_v2010.csv')
```

```
Rows: 57802 Columns: 48
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (12): provider_number, fy_start, fy_end, date_processed, date_created, d...
dbl (35): report, status, year, beds, tot_charges, tot_discounts, net_pat_re...
lgl  (1): npi

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
final.hcris.v1996 = final.hcris.v1996 %>%
  mutate(hvbp_payment=NA, hrrp_payment=NA)

final.hcris = bind_rows(final.hcris.v1996, final.hcris.v2010) %>%
  mutate(fy_end=mdy(fy_end), fy_start=mdy(fy_start),
         date_processed=mdy(date_processed), date_created=mdy(date_created),
         tot_discounts=abs(tot_discounts), hrrp_payment=abs(hrrp_payment)) %>%
  mutate(fyear=year(fy_end)) %>%
  arrange(provider_number, fyear) %>%
  select(-year)

q1 <- final.hcris %>%
  filter(fyear >= 2009, fyear <= 2019) %>%
  group_by(provider_number, fyear) %>%
  summarize(n_reports = n(), .groups = "drop") %>%
  filter(n_reports > 1) %>%
  group_by(fyear) %>%
  summarize(n_hospitals = n())

ggplot(q1, aes(x = fyear, y = n_hospitals)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "steelblue", size = 2) +
  labs(title = "Hospitals Filing More Than One Report Per Year",
       x = "Year", y = "Number of Hospitals") +
  theme_minimal()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



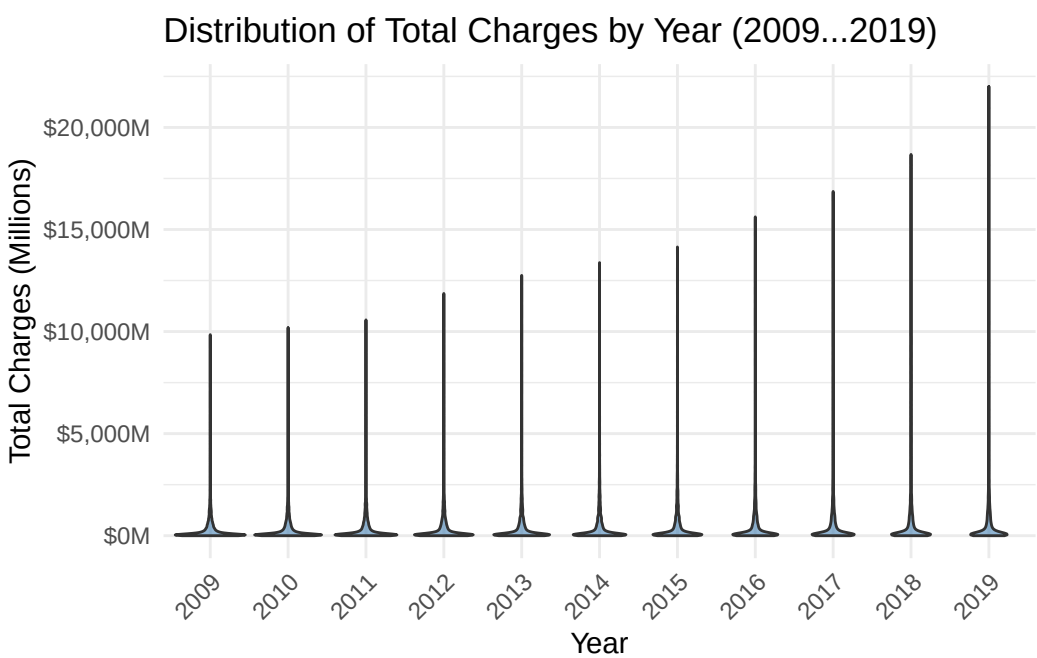

```
#Q2
n_unique <- final.hcris.data %>%
  distinct(provider_number) %>%
  nrow()

cat("Number of unique hospital IDs:", n_unique, "\n")
```

```
Number of unique hospital IDs: 9451
```

```
#Q3
q3 <- final.hcris.data %>%
  filter(year >= 2009, year <= 2019) %>%
  filter(!is.na(tot_charges), tot_charges > 0)

ggplot(q3, aes(x = factor(year), y = tot_charges)) +
  geom_violin(fill = "steelblue", alpha = 0.6) +
  scale_y_continuous(labels = scales::dollar_format(scale = 1e-6, suffix = "M")) +
  labs(title = "Distribution of Total Charges by Year (2009-2019)",
       x = "Year", y = "Total Charges (Millions)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
```

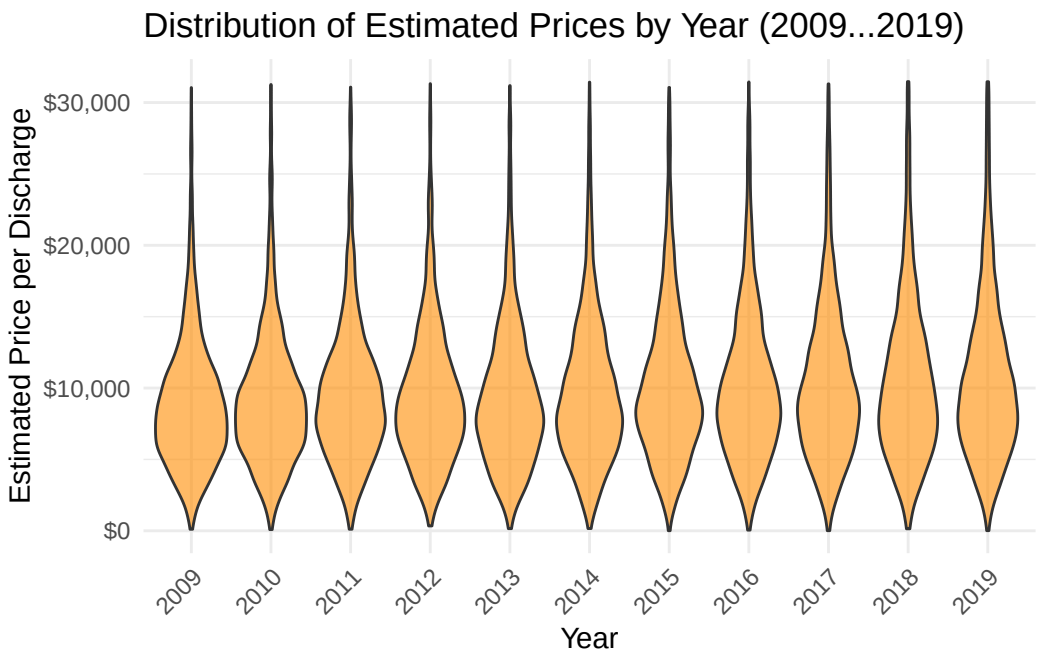
```
conversion failure on 'Distribution of Total Charges by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Total Charges by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Total Charges by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Total Charges by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Total Charges by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Total Charges by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <93>
```



```r
#Q4
q4 <- final.hcris.data %>%
  filter(year >= 2009, year <= 2019) %>%
  mutate(
    discount_factor = 1 - tot_discounts / tot_charges,
    price_num = (ip_charges + icu_charges + ancillary_charges) * discount_factor - tot_mcare
    price_denom = tot_discharges - mcare_discharges,
    price = price_num / price_denom
  ) %>%
  filter(
    !is.na(price),
    !is.infinite(price),
    price > 0,
    price < quantile(price, 0.99, na.rm = TRUE)
  )

ggplot(q4, aes(x = factor(year), y = price)) +
  geom_violin(fill = "darkorange", alpha = 0.6) +
  scale_y_continuous(labels = scales::dollar_format()) +
  labs(title = "Distribution of Estimated Prices by Year (2009-2019)",
       x = "Year", y = "Estimated Price per Discharge") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Estimated Prices by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Estimated Prices by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Estimated Prices by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Estimated Prices by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <e2>
```

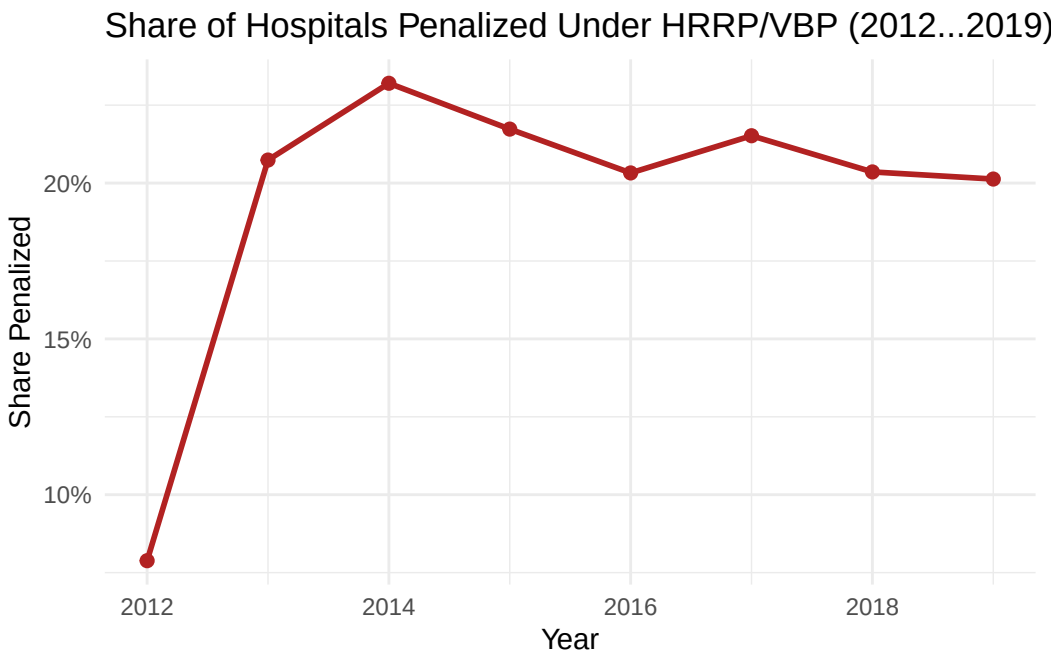
```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Estimated Prices by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Distribution of Estimated Prices by Year (2009-2019)' in
'mbcsToSbcs': dot substituted for <93>
```




```r
#Q5
q5 <- final.hcris.data %>%
  filter(year >= 2012, year <= 2019) %>%
  mutate(
    penalized = (!is.na(hrrp_payment) & hrrp_payment < 0) |
                (!is.na(hvbp_payment) & hvbp_payment < 0)
  ) %>%
  group_by(year) %>%
  summarize(share_penalized = mean(penalized, na.rm = TRUE))

ggplot(q5, aes(x = year, y = share_penalized)) +
  geom_line(color = "firebrick", size = 1) +
  geom_point(color = "firebrick", size = 2) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Share of Hospitals Penalized Under HRRP/VBP (2012-2019)",
```

```r
       x = "Year", y = "Share Penalized") +
  theme_minimal()
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Share of Hospitals Penalized Under HRRP/VBP (2012-2019)'
in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Share of Hospitals Penalized Under HRRP/VBP (2012-2019)'
in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Share of Hospitals Penalized Under HRRP/VBP (2012-2019)'
in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Share of Hospitals Penalized Under HRRP/VBP (2012-2019)'
in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Share of Hospitals Penalized Under HRRP/VBP (2012-2019)'
in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Share of Hospitals Penalized Under HRRP/VBP (2012-2019)'
in 'mbcsToSbcs': dot substituted for <93>
```




```
# Price calculation helper
calc_price <- function(df) {
  df %>%
    mutate(
      discount_factor = 1 - tot_discounts / tot_charges,
      price_num = (ip_charges + icu_charges + ancillary_charges) * discount_factor - tot_mca
      price_denom = tot_discharges - mcare_discharges,
      price = price_num / price_denom
    )
}

# Prices in 2011 and 2014
price_2011 <- final.hcris.data %>%
  filter(year == 2011) %>%
  calc_price() %>%
  select(provider_number, price_2011 = price)

price_2014 <- final.hcris.data %>%
  filter(year == 2014) %>%
  calc_price() %>%
  select(provider_number, price_2014 = price)

# 2012 net penalty
penalty_2012 <- final.hcris.data %>%
```

```r
  filter(year == 2012) %>%
  mutate(
    hrrp = replace_na(hrrp_payment, 0),
    hvbp = replace_na(hvbp_payment, 0),
    net_penalty = hrrp + hvbp,
    penalized = as.integer(net_penalty < 0)
  ) %>%
  select(provider_number, net_penalty, penalized)

# Pre-2012 averages (2009-2011)
pre2012 <- final.hcris.data %>%
  filter(year >= 2009, year <= 2011) %>%
  group_by(provider_number) %>%
  summarize(
    avg_mcare_discharges = mean(mcare_discharges, na.rm = TRUE),
    avg_beds = mean(beds, na.rm = TRUE),
    avg_mcaid_discharges = mean(mcaid_discharges, na.rm = TRUE)
  )

# Build analysis dataset
iv_data <- price_2011 %>%
  inner_join(price_2014, by = "provider_number") %>%
  inner_join(penalty_2012, by = "provider_number") %>%
  inner_join(pre2012, by = "provider_number") %>%
  mutate(price_change = price_2014 - price_2011) %>%
  filter(
    !is.na(price_change),
    !is.infinite(price_change),
    price_2011 > 0,
    price_2014 > 0,
    price_change > quantile(price_change, 0.01, na.rm = TRUE),
    price_change < quantile(price_change, 0.99, na.rm = TRUE)
  )

cat("Observations in IV dataset:", nrow(iv_data), "\n")
```

```
Observations in IV dataset: 2449
```

```r
#Q6
ols1 <- lm(price_change ~ net_penalty, data = iv_data)
ols2 <- lm(price_change ~ net_penalty + avg_beds, data = iv_data)
```

```r
ols3 <- lm(price_change ~ net_penalty + avg_beds + avg_mcaid_discharges, data = iv_data)

modelsummary(
  list("Baseline" = ols1, "+ Beds" = ols2, "+ Beds + Medicaid" = ols3),
  title = "OLS Estimates: Effect of Net Penalty on Price Change",
  stars = TRUE,
  gof_omit = "AIC|BIC|Log",
  output = "markdown"
)
```

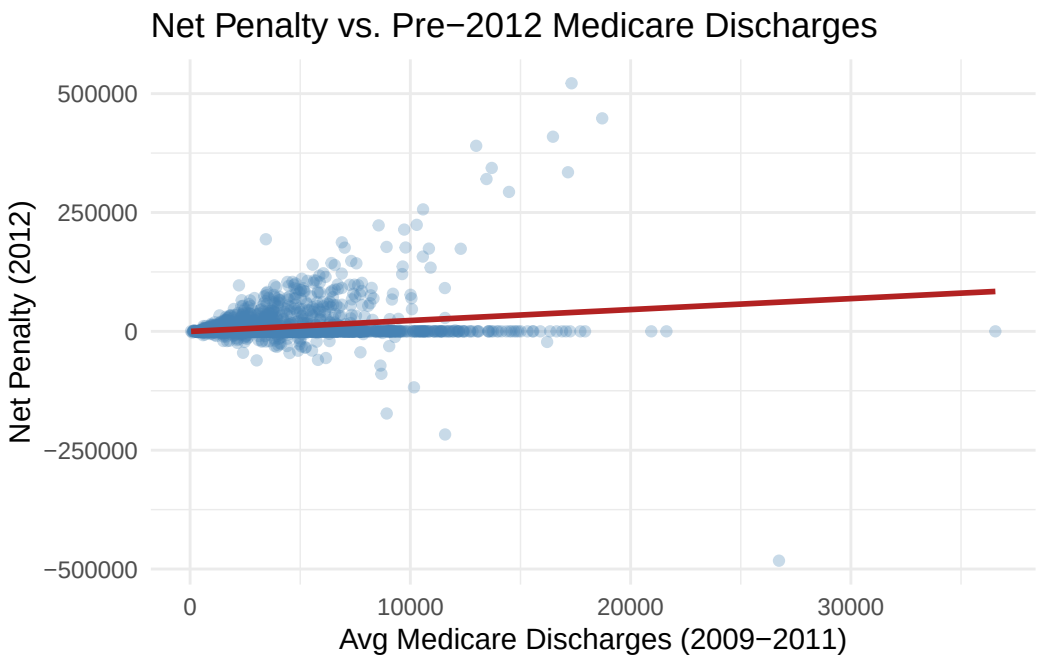
Table 1: OLS Estimates: Effect of Net Penalty on Price Change

	Baseline	+ Beds	+ Beds + Medicaid
(Intercept)	247.380***	229.061***	106.284+
	(47.055)	(47.565)	(58.872)
net_penalty	0.002	0.002	0.002
	(0.001)	(0.001)	(0.001)
avg_beds		0.074*	0.067*
		(0.030)	(0.030)
avg_mcaid_discharges			0.077***
			(0.022)
Num.Obs.	2449	2449	2440
R2	0.001	0.004	0.009
R2 Adj.	0.001	0.003	0.007
F	2.679	4.506	7.100
RMSE	2266.87	2263.94	2261.71

Note: ^^ + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```r
#Q7
ggplot(iv_data, aes(x = avg_mcare_discharges, y = net_penalty)) +
  geom_point(alpha = 0.3, color = "steelblue") +
  geom_smooth(method = "lm", color = "firebrick", se = FALSE) +
  labs(title = "Net Penalty vs. Pre-2012 Medicare Discharges",
       x = "Avg Medicare Discharges (2009-2011)",
       y = "Net Penalty (2012)") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
#Q8
fs1 <- lm(net_penalty ~ avg_mcare_discharges, data = iv_data)
fs2 <- lm(net_penalty ~ avg_mcare_discharges + avg_beds, data = iv_data)
fs3 <- lm(net_penalty ~ avg_mcare_discharges + avg_beds + avg_mcaid_discharges, data = iv_dat

rf1 <- lm(price_change ~ avg_mcare_discharges, data = iv_data)
rf2 <- lm(price_change ~ avg_mcare_discharges + avg_beds, data = iv_data)
rf3 <- lm(price_change ~ avg_mcare_discharges + avg_beds + avg_mcaid_discharges, data = iv_da

modelsummary(
  list("FS Baseline" = fs1, "FS + Beds" = fs2, "FS + Beds + Medicaid" = fs3,
       "RF Baseline" = rf1, "RF + Beds" = rf2, "RF + Beds + Medicaid" = rf3),
  title = "First Stage and Reduced Form Results",
  stars = TRUE,
  gof_omit = "AIC|BIC|Log",
  output = "markdown"
)
```

Table 2: First Stage and Reduced Form Results

	FS Baseline	FS + Beds	FS + Beds + Medicaid	RF Baseline	RF + Beds	RF + Beds + Medicaid
(Intercept)	-242.017	-232.107	295.239	168.136*	163.442*	126.855+
	(1046.415)	(1047.025)	(1061.451)	(70.135)	(70.097)	(71.048)
avg_mcare_discharges	2.303***	2.310***	2.717***	0.027+	0.023	-0.003
	(0.220)	(0.221)	(0.254)	(0.015)	(0.015)	(0.017)
avg_beds		-0.148	-0.110		0.070*	0.068*
		(0.443)	(0.443)		(0.030)	(0.030)
avg_mcaid_discharges			-1.232**			0.081**
			(0.377)			(0.025)
Num.Obs.	2449	2449	2440	2449	2449	2440
R2	0.043	0.043	0.047	0.001	0.004	0.008
R2 Adj.	0.043	0.042	0.046	0.001	0.003	0.007
F	109.866	54.969	40.085	3.319	4.457	6.406
RMSE	33817.26	33816.49	33803.99	2266.57	2263.99	2262.67

Note: ^^ + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
#Q9
iv1 <- ivreg(price_change ~ net_penalty | avg_mcare_discharges, data = iv_data)
iv2 <- ivreg(price_change ~ net_penalty + avg_beds | avg_mcare_discharges + avg_beds, data =
iv3 <- ivreg(price_change ~ net_penalty + avg_beds + avg_mcaid_discharges |
               avg_mcare_discharges + avg_beds + avg_mcaid_discharges, data = iv_data)

modelsummary(
  list("IV Baseline" = iv1, "IV + Beds" = iv2, "IV + Beds + Medicaid" = iv3),
  title = "IV Estimates: Effect of Net Penalty on Price Change",
  stars = TRUE,
  gof_omit = "AIC|BIC|Log",
  output = "markdown"
)
```

Table 3: IV Estimates: Effect of Net Penalty on Price Change

	IV Baseline	IV + Beds	IV + Beds + Medicaid
(Intercept)	170.955*	165.775*	127.194+
	(69.697)	(69.493)	(72.172)

	IV Baseline	IV + Beds	IV + Beds + Medicaid
net_penalty	0.012+	0.010	-0.001
	(0.006)	(0.006)	(0.006)
avg_beds		0.072*	0.067*
		(0.030)	(0.030)
avg_mcaid_discharges			0.079***
			(0.023)
Num.Obs.	2449	2449	2440
R2	-0.020	-0.011	0.006
R2 Adj.	-0.020	-0.012	0.005
RMSE	2290.43	2280.48	2264.20

Note: ^^ + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
#Q10
#The IV estimator identifies a **Local Average Treatment Effect (LATE)** which is the effect
#only for hospitals whose penalty status was driven by their pre-2012 Medicare discharge volu
#In this setting, compliers tend to be hospitals with higher Medicare exposure, which may res
#hospital, meaning the LATE could differ from the overall ATE. Our IV estimates (0.012 basel
#suggest the local effect among these Medicare-heavy hospitals is modest at best.
```